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Characteristics and sources of soil moisture dynamics in alpine meadows under simulated warming conditions

Zongjie Li [a,*], Hao Li [a], Bowen Wang [a], Juan Gui [b,**], Fei Xue [c], Xiaoying Liu [b], Fang Liu [a]

[a] School of Civil and Hydraulic Engineering, Lanzhou University of Technology, Lanzhou 730050, China
[b] Observation and Research Station of Eco-Hydrology and National Park by Stable Isotope Tracing in Qilian Mountains/National Key Laboratory of Ecological Security and Sustainable Development in Arid Region, Northwest Institute of Eco-Environment and Resources, Chinese Academy of Sciences, Lanzhou 730000, China
[c] Hydrology and Water Resources Survey Center in Hanzhong, Hanzhong 723000, China

A R T I C L E I N F O

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A B S T R A C T

This study quantified the dynamics and sources of soil moisture in alpine meadows under experimental warming during 2023–2024 at the observation and research Station of eco-hydrology and national park by stable isotope tracing in Qilian Mountains. Except for the shallow layer (0–10 cm) under warming plus increased precipitation (ZWZY), overall soil water content in 2024 exceeded that in 2023; in 2023, all depths in the warming treatment (ZWCK) were drier than the control (CKCK). Soil-water isotopes showed evaporative enrichment from June ($\delta^{18}O \approx -9.88$ ‰; $\delta^{2}H \approx -77.41$ ‰) to the peak-heat period in July (≈ -6.70 ‰; ≈ -40.46 ‰), followed by monsoon-driven depletion and isotopic convergence in August. The summer soil–water line slope (≈ 6.64) was lower than the local meteoric water line (≈ 7.71), indicating non-equilibrium evaporation, whereas rapid post-storm isotope shifts between July and October point to precipitation-pulse-dominated recharge. Among environmental controls, soil temperature was the primary driver of isotopic variability, with precipitation as a secondary control. Mechanistically, piston flow contributed ≥ 85 % in June–July and October, while heavy rainfall increased preferential flow to 38–48 % in August–September. We propose a warming-responsive restoration framework for alpine meadows: canopy retention to damp heat load, three-tier water management (surface–root zone–subsoil), and targeted night-time water supplementation to buffer future climate uncertainty.

1. Introduction

Over the past several decades, the Qinghai-Tibet Plateau region has experienced unprecedented changes, including rapid climate change and significant warming (Chen et al., 2013; Yao et al., 2019). Cold-region ecosystems exhibit high sensitivity to temperature changes, particularly in the cold-temperate meadow soils of high-altitude regions such as the Qinghai-Tibet Plateau (Cao et al., 2015). The cascading responses of hydrological processes—including changes in water dynamics, restructuring of precipitation spatiotemporal patterns, and enhanced potential evapotranspiration—driven by warming have become one of the core issues in current ecological hydrology research. Preventing the degradation of the Shaliu River's water-conservation function and ensuring effective management and protection of alpine-meadow soil water resources remain critical scientific challenges. Beyond the Tibetan Plateau, a three-year isotope–hydrology study in the Alps clarified dominant runoff mechanisms in small catchments during rainfall and snowmelt, highlighting rapid event-scale responses and substantial pre-event water contributions—providing a useful benchmark for cross-mountain comparisons of hydroclimatic coupling (Penna et al., 2016).

Soil water infiltration behavior is a key component of surface hydrological processes, and its dynamic characteristics directly influence water conversion and distribution at the watershed scale. Therefore, accurately characterizing soil infiltration mechanisms is of great significance for comprehensively understanding regional hydrological cycles (Jiao et al., 2020). Research indicates that soil moisture is not only a key variable linking atmospheric, hydrological, and biological processes

* Corresponding author at: School of Civil and Hydraulic Engineering, Lanzhou University of Technology, Lanzhou 730050, China.
** Corresponding author at: Observation and Research Station of Eco-Hydrology and National Park by Stable Isotope Tracing in Qilian Mountains/National Key Laboratory of Ecological Security and Sustainable Development in Arid Region, Northwest Institute of Eco-Environment and Resources, Chinese Academy of Sciences, Lanzhou 730000, China.
E-mail addresses: lzjie314@163.com (Z. Li), guijuan6894@163.com (J. Gui).

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but also the primary factor limiting vegetation productivity and carbon cycling in cold regions. As one of the primary cold-temperate ecosystems, the dynamic changes in soil moisture in cold-temperate meadows are primarily controlled by multiple factors such as soil structure, precipitation characteristics, and climate change, and their water source composition often exhibits significant spatiotemporal variability (Zong et al., 2018; Chen et al., 2013).

In the tropical Andes páramo, EMMA combined with isotopic tracers shows that organic-rich riparian wet soils dominate year-round streamflow contributions, with spring inputs rising in the dry season (Correa et al., 2017). This mirrors the "shallow mixing–rapid convergence" pattern in alpine meadows and suggests comparable soil–vegetation–hydrology control points across climates and soil types.

Stable isotope techniques, with their temporal integration and source-tracing capabilities, help overcome limitations of traditional hydrological observations and have been widely applied in cold-region hydrology (Sprenger et al., 2016; McDonnell & Beven, 2014). Using these approaches, Gui et al. (2023) quantitatively identified and analyzed the sources of soil water and their controlling factors in the Qilian Mountains on the northeastern edge of the Tibetan Plateau. Similarly, by systematically collecting soil water, precipitation, river water, surface ice, upper permafrost water, and glacial snowmelt water, Li et al. (2024c) quantified the relative contributions of multiple water bodies to soil–water replenishment across different melt periods in 2020 in the headwaters of the Three Rivers.

Beyond source apportionment, stable-isotope tracing with two-endmember mixing has illuminated infiltration mechanisms and related hydrological processes. In wetlands, Cong et al. (2023) showed spatiotemporal variation in plant water sources and soil-moisture stratification, underscoring vegetation's role in the water cycle; in alpine meadows. Zongjie et al. (2025) resolved the primary sources and pathways of soil–water recharge, revealing layer-specific and seasonal contributions from precipitation and springs. Identifying dominant soil–water sources is therefore essential for science-based water management and ecological protection. At high latitudes, isotope observations from Alaska's Barrow polygonal permafrost indicate summer rainfall dominates active-layer soil water, with late-season contributions from seasonal ice melt and pervasive non-equilibrium evaporative fractionation in surface waters—necessitating separation of evaporation–freezing effects (Throckmorton et al., 2016). In the West Siberian Lowlands, a multi-year, 1,700-km transect shows spring snowmelt mixes with existing soil/lake storage and displaces older water to generate runoff, while wetland–permafrost mosaics sustain high connectivity during ablation, controlling river-network convergence and source-area expansion (Ala-Aho et al., 2018). Together, these studies provide a template for assessing cold-region source composition, connectivity, and warming responses.

In recent years, LC-excess (line-conditioned excess) has emerged as a relatively new stable isotope indicator increasingly used to identify whether water bodies have undergone non-equilibrium evaporation processes. Its combination with the end-member mixing model (EMMA) can effectively quantify the relative contributions of different water sources to soil water (Zhong et al., 2023; Sprenger et al., 2017). Additionally, the theoretical model combining piston flow and preferential flow has become an effective tool for analyzing the vertical migration pathways of soil water (Huang et al., 2020; Weiler & McDonnell, 2004). This study comprehensively incorporates the aforementioned methods to systematically analyze the sources and infiltration mechanisms of shallow (0–10 cm) and deep (10–20 cm) soil water in a typical alpine meadow area of the Shaliu River Basin in Qinghai Lake, exploring structural changes in hydrological processes under different warming scenarios. Cross-regional evidence also shows that soil–water stable isotope profiles are increasingly used to identify preferential flow in alpine catchments (e.g., the Alps), enabling the localization of multiple vertically discontinuous fast pathways at hillslope–small-catchment scales and improving structural representations of infiltration and runoff generation (Pyschik & Weiler, 2025).

This study is to our knowledge the first to introduce the LC-Excess and EMMA models in the Shaliu River basin of Qinghai Lake, systematically and quantitatively assessing the proportion of groundwater replenishment from subglacial ice, permafrost surface water, and precipitation. By constructing a dual-mode model of piston flow and preferential flow, the study revealed the dominant vertical flow pathways of alpine meadow soil under warming scenarios. Combining continuous isotope observation data from 2023 to 2024, the study deepens our understanding of how warming driven by interannual climate variability affects soil hydrological processes. The findings will provide scientific basis and decision-making references for ecological conservation and water resource management in the Qinghai-Tibet Plateau and similar cold regions.

2. Materials and methods

2.1. Study area

The experimental site is situated on the northeastern edge of the Qinghai–Tibet Plateau (37° 46′ 48″ N, 99° 42′ 00″ E) at the he observation and research Station of eco-hydrology and national park by stable isotope tracing in Qilian Mountains, Ministry of Ecology and Environment of China. It lies within the Shaliu River basin in northern Gangcha County, Qinghai Province (Fig. 1), an important recharge tributary of the Qinghai Lake basin. The river originates from the southern slope of Mount Sangsizha, the highest peak in Gangcha County, with an elevation of approximately 5,075 m, and is part of the alpine mountainous river system (Li et al., 2010). The Shaliu River, with a total length of about 106 km, contributes approximately one-fifth of the total inflow to Qinghai Lake, ranking second in the basin. The region has an alpine semi-arid climate, with a mean annual temperature of ~ 0.1 °C, annual precipitation of ~ 400 mm, and potential evaporation reaching up to 1,300 mm. Most evapotranspiration occurs during the warm summer months. Dominated alternately by the East Asian monsoon and westerly circulation, the region exhibits strong seasonal and spatial variability in hydrothermal distribution (Yang et al., 2020), typical of a plateau continental climate (Gangcha County Annals Compilation Committee, 1998), with long sunshine hours and large diurnal temperature fluctuations.

2.2. Sampling design and experimental design

We employed a randomized block design with two factors and their combinations (Fig. 2): precipitation at three levels (−30 %, unchanged, +30 %) and temperature at two levels (ambient, elevated). Each treatment had five replicates, yielding 30 plots (Table 1) (3 precipitation × 2 temperature × 5 replicates). Plots measured 3 × 3 m with 2–3 m spacing. Warming was applied using hexagonal transparent acrylic open-top chambers (OTCs) with a 60° inward-sloping roof; each OTC was 60 cm wide at the top, 106 cm at the base, and 40 cm high, producing an average warming of + 2 °C (The sensor is from METER Corporation, USA, model ATMOS14). Rain-reduction frames (3 × 3 m) comprised multiple inclined V-shaped acrylic panels mounted 1 m aboveground on metal supports, covering 30 % of the plot surface. Intercepted rainfall was collected in containers and, within 24–48 h after each event, manually transferred to the nearest rain-enhancement plot, thereby increasing its natural precipitation by 30 % without changing event frequency. During the growing season, 5–7 such transfers were conducted per month.

Soil samples (n = 108): During the 2023–2024 growing seasons, 36 sites were sampled at monthly intervals (end of month). Measured soil gravel content ranged from 1 % to 18 %. At each site, a 20-cm soil core was extracted from alpine-meadow soil using a 7-cm-diameter auger and partitioned into two layers: 0–10 cm and 10–20 cm. Subsamples were immediately sealed in pre-cleaned 100-mL HDPE bottles, wrapped with

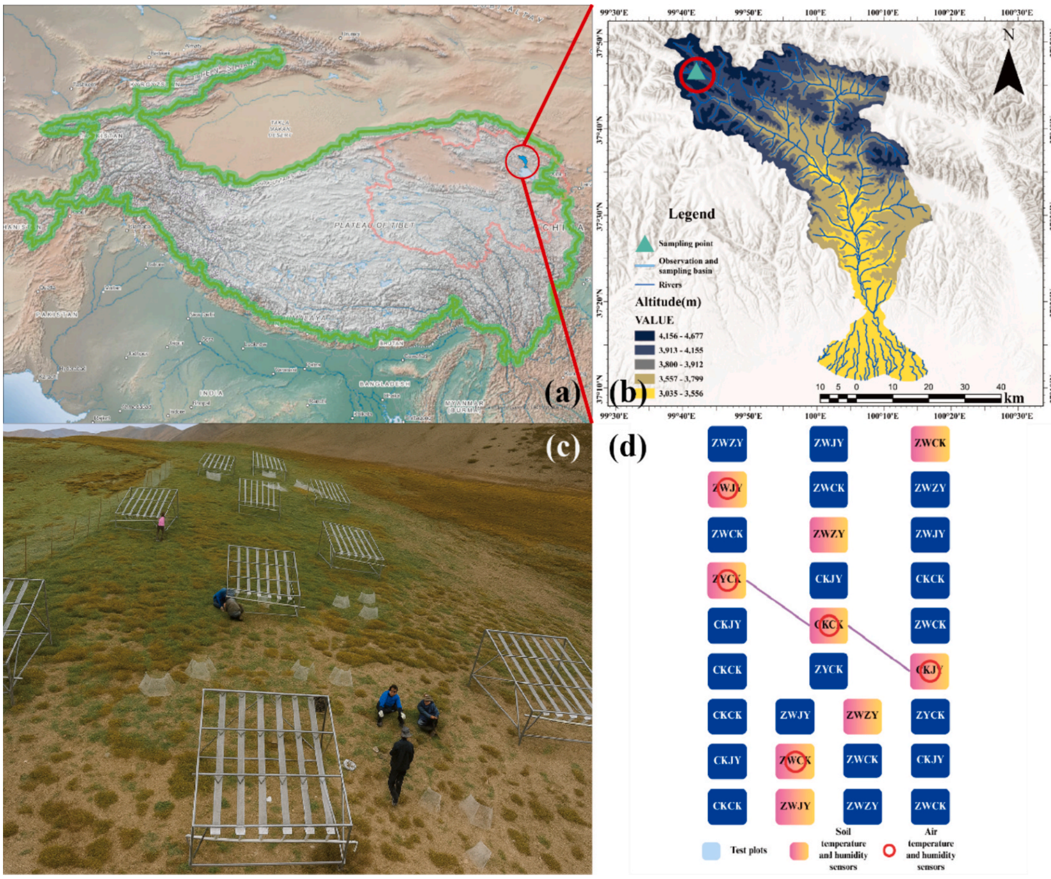

Fig. 1. a-d show the location of the site on the Qinghai-Tibet Plateau, the specific location of the experimental area, a physical map of the experimental area, and a schematic diagram of the field experiment layout for the warming-precipitation gradient scenario, respectively.

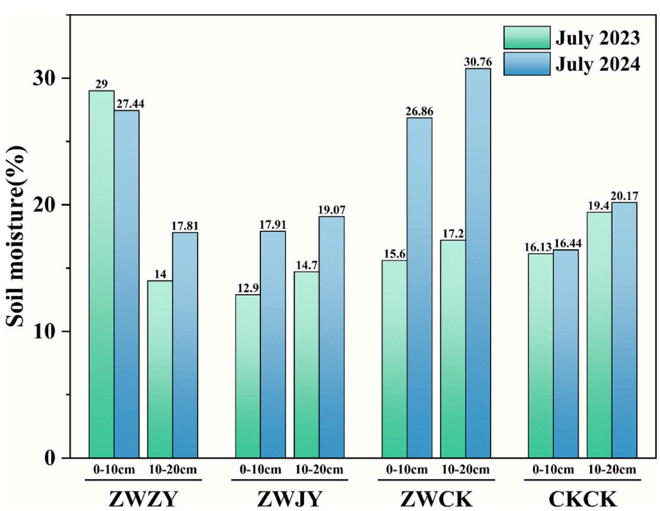

Fig. 2. Variation in soil moisture content under warming treatments in July of 2023 and 2024 (ZWZY: warming + increased precipitation, ZWJY: warming + decreased precipitation, ZWCK: warming + control check, and CKCK: control check).

Parafilm to minimize moisture loss, and stored at low temperature until analysis. Concurrently, in situ volumetric water content and temperature were recorded using a portable sensor (TZS-IW). During sampling, the dominant plants within the OTCs were *Artemisia alpinis* Willd. and *Urtica hyperborea* Jacq. ex Wedd.

Precipitation samples (n = 194): Event-based samples were collected between 2020 and 2022 in the Shaliu River Basin using standard rainfall recorders, then transferred into pre-cleaned 50 ml HDPE bottles, sealed with Parafilm, and sent to the laboratory for analysis.

Ground ice samples (n = 41): Sourced from published data (Gui et al., 2023), extracted from the permafrost active layer in the Qilian Mountains. Ice samples were obtained by excavating 1-meter profiles, with surface contaminants removed. Crushed ice was sealed in HDPE bottles, Parafilmed, and frozen.

Supra-permafrost water samples (n = 25): Collected across different thawing stages over two years in the Shaliu River Basin. Samples were extracted from ~ 2 m deep pits, filtered on-site through 0.45 μm membranes. Bottles were pre-rinsed with target water, sealed with Parafilm, and transported under refrigerated conditions.

Table 1
Treatment codes and corresponding conditions.

Code	Treatment	Code	Treatment
ZWZY	+2 °C warming & +30 % precipitation	CKZY	+30 % precipitation
ZWJY	+2 °C warming & –30 % precipitation	CKJY	–30 % precipitation
ZWCK	+2 °C warming	CKCK	Control (no treatment)

2.3. End-Member mixing analysis (EMMA)

The end-member mixing analysis (EMMA) is a widely used approach to assess the contributions of various water sources (Hooper, 2003; Li et al., 2016a, Li et al., 2016b). A three-end-member model was applied to calculate the relative contributions of precipitation, supra-permafrost water, and ground ice using the following equations:

$$X_s = F_1X_1 + F_2X_2 + F_3X_3 \tag{1a}$$

$$Y_s = F_1Y_1 + F_2Y_2 + F_3Y_3 \tag{1b}$$

where X and Y represent tracer concentrations (here, d-excess and δ^{18}O), and subscripts 1, 2, and 3 denote the three major end-members. F indicates the fractional contribution of each source. When three sources are assumed to control the target water composition, the contributions can be calculated using equations (2a), (2b), and (2c):

$$F_1 = \frac{\frac{(X_3-X_s)}{(X_3-X_2)} - \frac{(Y_3-Y_s)}{(Y_3-Y_2)}}{\frac{(Y_1-Y_3)}{(Y_3-Y_2)} - \frac{(X_1-X_3)}{(X_3-X_2)}} \tag{2a}$$

$$F_2 = \frac{\frac{(X_3-X_S)}{(X_3-X_1)} - \frac{(Y_3-Y_S)}{(Y_3-Y_1)}}{\frac{(Y_2-Y_3)}{(Y_3-Y_1)} - \frac{(X_2-X_3)}{(X_3-X_1)}} \tag{2b}$$

F_1 and F_2 are the contribution rates of different water sources, which can be calculated using the following formula (2c):

$$F_3 = 1 - F_1 - F_2 \tag{2c}$$

2.4. Lc-excess method

The line-conditioned excess (lc-excess) method (Zhong et al., 2023) quantitatively expresses the relationship between δ^2H and δ^{18}O values. When water undergoes isotopic equilibrium processes such as condensation, lc-excess is typically positive; under non-equilibrium processes like evaporation, lc-excess tends to be negative. Local precipitation generally has an lc-excess near zero. The calculation is as follows:

$$lc - excess = \delta^2H - \alpha\delta^{18}O - b \tag{3}$$

Where: α is the slope of the local meteoric water line (LMWL); b is the intercept of the LMWL. The LMWL is fitted to the hydrogen–oxygen stable isotope values of the local precipitation:

$$\delta^2H = (7.85 \pm 0.14)\delta^{18}O + (21.88 \pm 1.67)(R^2 = 0.88, P < 0.05)$$

2.5. Lc-excess balanced equation

Given that piston flow and preferential flow often produce distinct isotope signals in soil water, a two-end-member isotopic mass balance was employed to quantify their respective contributions:

$$f_{PF} = \frac{lc_{DSW} - lc_{pf}}{lc_{PF} - lc_{pf}} \tag{4}$$

Where: f_{PF} is the relative contribution rate of the piston flow pattern, and the contribution rate of the preferential flow can be calculated from $f_{PF}+f_{pf} = 1$. lc_{DSW} selects the lc-excess value of soil water at 100 ~ 120 cm in the vertical soil profile; lc_{pf} selects the lc − excess value of precipitation in the same month in which the signal of preferential flow is obviously appeared; lc_{PF} is the lc-excess value of soil water at the sampling point where the piston flow signal appears (Zhong et al., 2023).

3. Results

3.1. Variation characteristics of soil moisture content

Based on observational data collected during July in both 2023 and 2024, this study analyzed the moisture-response patterns of the different treatment groups (ZWZY, ZWJY, ZWCK, CKCK) in the 0–10 cm and 10–20 cm soil layers (Fig. 2). The moisture responses across treatment groups revealed that, under warming conditions—and except for the shallow soil in ZWZY—water content in 2024 was significantly higher than in 2023. Comparing the 2023 and 2024 data, the mean moisture of all layers in the four experimental groups rose from 17.39 % to 22.06 %, an increase of 4.67 %. In the shallow layer, moisture climbed from 18.41 % to 22.16 %, whereas the deeper layer fluctuated more markedly (16.33 % → 21.95 %).

Second, in 2023, water content in every layer of ZWCK was lower than in CKCK, but in 2024 it surpassed CKCK by a substantial margin—likely because measurements were taken just after rainfall, inflating the water-content values.

In ZWZY, shallow-layer moisture (2023 vs 2024) always exceeded that of the 10–20 cm layer: 0–10 cm recorded 29 % and 27.44 %, while 10–20 cm showed only 14 % and 17.81 %. This suggests that, after artificial warming and increased precipitation, rainwater had not yet infiltrated to depth, leaving deeper soil comparatively drier. Warming may have intensified the vertical gradient by enhancing evaporation from the surface and root water uptake.

In ZWJY, shallow-layer moisture in 2023 (12.9 %, 14.7 %) was markedly lower than in CKCK (16.13 %, 19.4 %), and deep-layer moisture in 2024 was likewise lower than CKCK (19.07 % → 20.17 %), indicating that reduced-rainfall treatments may have diminished soil water retention.

3.2. Stable isotope characteristics of soil water

Fig. 3 illustrates the monthly variations in δ^{18}O, d-excess, and δ^2H of soil water (0–10 cm depth) during June–October 2023 and July–September 2024. The δ^{18}O values ranged from − 10.99 ‰ to − 6.28 ‰, δ^2H ranged from − 85.80 ‰ to − 44.17 ‰, and d-excess varied from − 9.99 ‰ to + 17.71 ‰. These results reflect the alternating influence of evaporation and precipitation on soil water in the alpine meadow during the growing season.

In June, δ^{18}O and δ^2H exhibited their largest variability (−9.88 ‰, −77.41 ‰), reflecting the combined influence of intermittent rainfall and strong surface evaporation during the dry season. In July, under high temperatures, both isotopes became markedly enriched (−6.70 ‰, −40.46 ‰), indicating intense evaporative fractionation. In August, East Asian monsoon rainfall drove isotopic depletion and convergence. By September, moderate enrichment reappeared, while in October, declining temperatures and sharply reduced rainfall caused δ^{18}O to rise again, though with a much narrower range (−9.90 ‰, −66.21 ‰).

Seasonal patterns of d-excess provided further insights into moisture sources and redistribution processes. During June–July, average d-excess values ranged from 1 ‰ to 13 ‰, significantly lower than the global land average (10 ‰), supporting the inference that evaporation dominated the isotopic composition of soil water, especially when combined with δ^{18}O enrichment. From August to September, d-excess increased to 5–13 ‰, and both δ^{18}O and δ^2H values moved closer to the Global Meteoric Water Line (GMWL), indicating rapid infiltration of fresh, low-temperature monsoonal precipitation that diluted the previously evaporated water. By October, d-excess dropped below 0 ‰, suggesting a shift back to evaporation dominance as precipitation diminished.

The seasonal dynamics shown in Fig. 3 clearly demonstrate the isotopic fractionation and mixing processes of soil water under alternating evaporation and monsoon-driven recharge. These findings offer key quantitative reference parameters for future stable isotope modeling and ecohydrological forecasting on the Qinghai–Tibet Plateau under climate warming scenarios.

3.3. Soil water line (SWL)

In this study, the local meteoric water line (LMWL) was derived from 182 precipitation samples collected between 2023 and 2024 in the study area, yielding the regression equation: δ^2H = 7.71 δ^{18}O + 19.98 (R^2 = 0.93). Compared to the Global Meteoric Water Line (GMWL: δ^2H = 8 δ^{18}O + 10), the LMWL had a slightly lower slope and a significantly higher intercept, indicating partial sub-cloud evaporation during precipitation transport. In addition, the elevated intercept reflects the

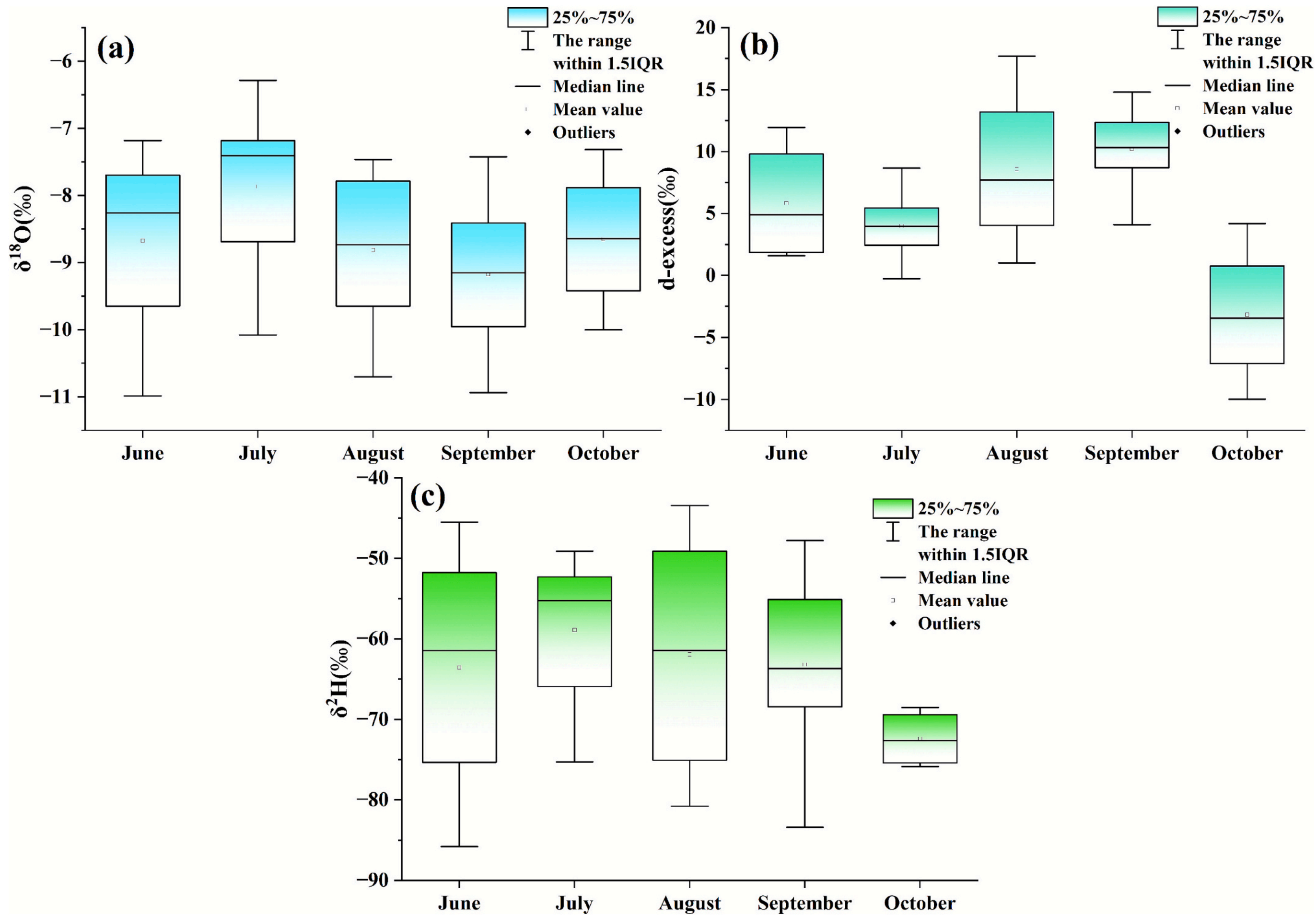

Fig. 3. Variations of soil water δ^2H, δ^{18}O, and d-excess values during the growing season from June to October in 2023 and 2024.

influence of high-elevation and monsoon-derived vapor, which typically contains higher d-excess. The high R^2 value confirms the strong seasonal correlation between δ^{18}O and δ^2H, providing a robust baseline for subsequent evaluations of evaporative fractionation and water mixing processes.

Observing Fig. 4a, it is evident that the soil water line (SWL) in 2023 exhibits significant inter-monthly variations. The regression slopes for June–September (5.83, 6.64, 4.92, 5.20) are all significantly lower than the local atmospheric water line (LMWL) of 7.71, while the October slope rises to 7.70, nearly overlapping with the LMWL, indicating that soil water in summer is controlled by continuous evaporative fractionation, while in autumn it tends toward the atmospheric water line due to rapid replenishment from new precipitation. The slope in August is the lowest and the intercept (−20.93 ‰) is the most negative, indicating the

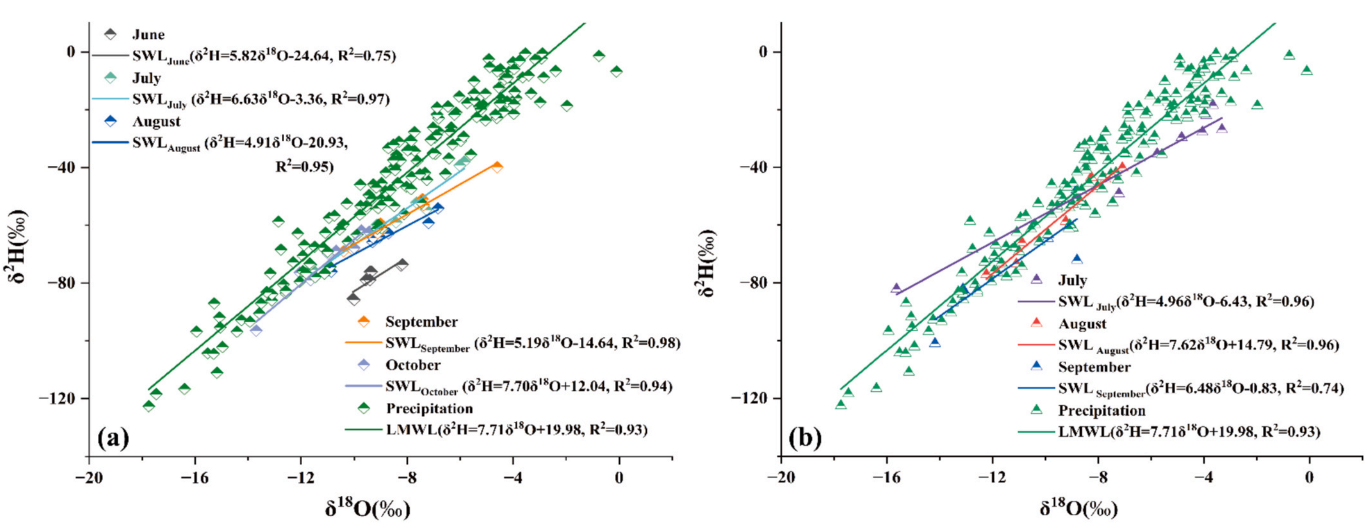

Fig. 4. Soil water lines (SWL) during June–October 2023 (a) and July–September 2024 (b).

strongest evapotranspiration during the high-temperature period. This highly negative intercept reflects strongly evaporated soil water, consistent with the regression plotting far below the LMWL. Although evaporation is vigorous in July, the intercept is only − 3.36 ‰, suggesting that heavy precipitation produces a pronounced “dilution” effect that offsets evaporative enrichment. In September, as temperatures decline, the slope and intercept (5.19, −14.64 ‰) fall between those of midsummer and autumn. In October, the intercept turns positive (+12.04 ‰), coupled with a high slope, revealing that precipitation infiltration dominates and evaporation fractionation weakens. Overall, the seasonal deviation of SWL can be used to quantitatively estimate the evaporation coefficient and precipitation recharge ratio, providing monthly-scale constraints for water–air-soil interactions and model parameterization in alpine meadow ecosystems.

The R^2 values for July and August 2024 were 0.96, indicating a very strong linear relationship, suggesting that the changes in δ^2H and δ^{18}O values of soil moisture in these two months showed good linear fitting (Fig. 4b). However, the R^2 value for September was lower at only 0.74, indicating that the correlation between δ^2H and δ^{18}O in soil water during September was less robust than in July and August. This may be attributed to soil water mixing and delayed infiltration caused by soil heterogeneity/preferential flow (Allison and Hughes, 1983; Sprenger et al., 2016; Gazis and Feng, 2004). The slope for August (7.62) is the highest among the three seasons, indicating that δ^2H is more sensitive to changes in δ^{18}O during August. The slope of the soil water line for July is only 4.96, significantly lower than the local atmospheric water line (7.71), and the intercept is negative, indicating that evaporation fractionation is strongest during the high-temperature phase, with heavy isotopes highly enriched in the liquid phase; in August, the slope rises to 7.62 and the intercept to + 14.79, nearly overlapping with the atmospheric water line, reflecting the extensive infiltration of monsoon precipitation, diluting the earlier evaporation signals; in September, the slope dropped back to 6.48 (intercept − 0.83, $R^2 = 0.74$), with evaporation once again dominating, but due to lower temperatures and intermittent precipitation, the fractionation amplitude was smaller than during the peak summer.

The relationship between soil water isotopes (δ^2H and δ^{18}O) in July, August, and September exhibits significant seasonal variations. The changes in August are the most pronounced, indicating that climatic conditions (such as precipitation and evaporation) during this month have a significant impact on the isotopic composition of soil water. The lower R^2 value in September suggests that soil water changes during this month may be influenced by other factors.

4. Discussions

4.1. Effects of warming on stable isotopes in soil water

The stable isotope composition of soil water (δ^2H and δ^{18}O) is jointly regulated by multiple environmental factors, primarily including the coupled effects of soil temperature, humidity, bulk density, and vegetation cover (Sprenger et al., 2016; Geris et al., 2015). Soil temperature is the key factor driving soil water evaporation fractionation. Under high-temperature conditions, the lighter H and ^{16}O isotopes in water molecules are more easily volatilized, leading to the enrichment of heavier isotopes in residual soil water (Allison & Hughes, 1983; Penna et al., 2018).

The stable isotopes of soil water in alpine meadows exhibit consistent seasonal variations under the coupled effects of warming and moisture (Fig. 5). In June, the δ^2H value for all treatments was approximately − 80 ‰, the δ^{18}O value was approximately − 10 ‰, and the d-excess value was approximately 0 ‰, representing the isotopic baseline before the rainy season. In July, when temperatures peaked, evaporative fractionation became dominant, with the control group showing δ^2H enrichment to − 35 ‰, δ^{18}O to − 4.5 ‰, and d-excess increasing to 8 ‰ (Fig. 5a-d). However, as the warming gradient increased, the median values instead decrease (δ^2H ≈ −55 to − 60 ‰, δ^{18}O ≈ −7.5 to − 2 ‰), and the variability and d-excess (12 to 18 ‰) significantly increase, indicating that although warming intensifies evapotranspiration, the decrease in water content and the triggering of preferential flow lead to

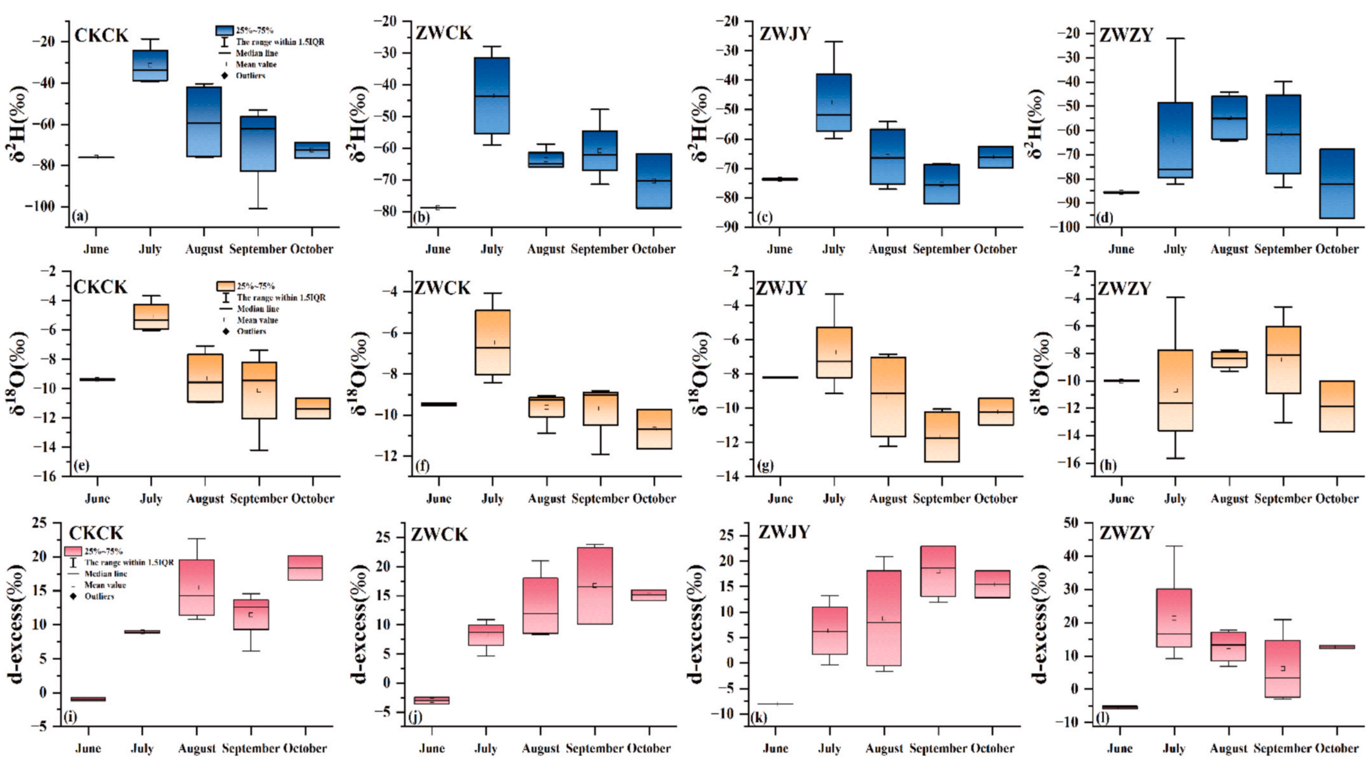

Fig. 5. Changes in soil isotopes under warming across soil types: δ^2H (a–d), δ^{18}O (e–h), and d-excess (i–l), (CKCK: control check, ZWCK: warming + control check, ZWJY: warming + decreased precipitation, and ZWZY: warming + increased precipitation).

frequent mixing with isotopically depleted water, offsetting part of the enrichment (Sprenger et al., 2017). Monsoon precipitation in August–September dilutes the overall isotopic composition (both δ^2H and δ^{18}O decline). Notably, the warming zone remains 5–15 ‰ more negative—i.e., more depleted, paradoxically opposite to the evaporative enrichment usually expected—reflecting greater preferential flow of isotopically light rainfall water. The simultaneous expansion of the interquartile range (IQR) indicates that spatial differences in evaporation–recharge competition are amplified under high-temperature, low-humidity conditions. At this time, δ excess surges across all treatments (15–25 ‰), indicating increased new water content and highly heterogeneous infiltration–evaporation responses. In October, precipitation sharply decreases and temperatures drop, isotope values slightly enriched and converged, but the warming group still maintained negative deviations in δ^2H and δ^{18}O and higher d-excess (Zhong et al., 2023), reflecting that residual water bodies in late autumn were dominated by enriched "old water" and still retained the memory of warming. Overall, δ^{18}O shows a larger response amplitude to warming-moisture interactions than δ^2H, while d-excess links the dual effects of evaporation fractionation and new water mixing. Together, these three factors indicate that warming significantly amplifies the spatiotemporal heterogeneity of soil water isotopes through a coupled mechanism of "enhanced evapotranspiration + reduced water content + triggering preferential flow" (Fig. 5e-l), and highlights the key regulatory role of rapid precipitation replenishment in the water cycle of alpine meadows.

The stable isotope variations in soil water of alpine meadow soils exhibit a distinct "primary–secondary" driving pattern: soil temperature is the primary controlling factor. In July, under the warming treatment (ZWZY), δ^{18}O enriched from − 10 ‰ at the beginning of the season to − 2 ‰, an enrichment magnitude approximately 80 % higher than that of the control CKCK treatment (−10 → − 4.5 ‰). Concurrently, the peak d-excess value increased from 8 ‰ to 18 ‰, indicating that warming significantly enhanced evaporation fractionation and accelerated the dynamic cycle of "old vs. new water." Soil moisture content and precipitation replenishment are key secondary regulators. During the monsoon peak (August–September), δ^2H and δ^{18}O values for all treatments decreased in the negative direction, but the warming group remained 5–15 ‰ lower than CKCK, with significantly increased variability. This suggests that under high-temperature, low-humidity conditions, the dilution effect of precipitation is limited, amplifying the evaporation signal. Pore structure and vegetation root systems further refine this process by altering water migration pathways—high-density soils have low infiltration capacity and restricted preferential flow, preserving evaporation signals; while loose soils or dense root systems are prone to preferential flow, weakening evaporation enrichment. At the interannual scale, in 2023, under moist conditions, the surface δ values were generally negative and d-excess was positive, indicating that precipitation dominated; in 2024, increased aridity was characterized by heavy isotope enrichment and d-excess was often negative, consistent with the regional water cycle trend of "precipitation concentration—evaporation intensification—complex infiltration pathways." In summary, soil temperature dominates isotope enrichment intensity through the evaporation fractionation mechanism, while soil moisture conditions regulate temperature signals by controlling dilution and mixing processes. Pore characteristics, vegetation, and interannual climate jointly shape the details of isotope spatiotemporal heterogeneity.

In addition to soil temperature and moisture content, soil bulk density is another influencing factor. Soils with higher bulk density have lower porosity and weaker water retention capacity, leading to strong evaporation and insufficient replenishment in shallow water bodies. In contrast, soils with lower bulk density and looser texture are more prone to preferential flow, reducing the evaporation enrichment effect (Cheng & Jin, 2013). The water uptake by plant roots may also alter the isotopic composition of soil water. Studies have shown that while plant transpiration does not cause fractionation of water, root systems can preferentially absorb water from specific layers, thereby indirectly influencing the isotopic distribution of residual water bodies (Dawson & Ehleringer, 1991; McDonnell & Beven, 2014). In addition to natural factors, interannual climate variability is also an important background variable driving changes in soil water isotopes. 2023 was a relatively wet year, with overall negative surface isotopes and positive d-excess values, indicating that precipitation dominated; whereas in 2024, drought intensified, with evaporation fractionation dominating the evolution of water bodies. This interannual variation aligns closely with the recent trend in the water cycle on the Tibetan Plateau, characterized by "precipitation intensification—evaporation enhancement—complexification of infiltration pathways" (Yao et al., 2012).

4.2. *Effects of warming on infiltration process inferred from lc-excess*

Under conditions dominated by piston flow, soil water exhibits a layered downward movement, with newly infiltrated water gradually replacing older water in deeper layers, thereby enhancing the mixing between different water layers (Zimmermann et al., 1966). In contrast, when preferential flow is the primary migration mechanism, water can rapidly infiltrate through large pores or cracks in the soil, bypassing part of the soil matrix layer, thereby limiting mixing between water bodies of different ages (Beven and Germann, 2013). Therefore, when the stable isotope composition of soil water is close to that of precipitation, it typically indicates that it has undergone minimal mixing; conversely, a significant isotopic deviation between the two suggests that younger water has undergone more thorough mixing with the original water in the soil during infiltration (Du et al., 2024).

Combined evidence from δ^2H–δ^{18}O dual isotope regression and lc-excess vertical sequences indicates that soil moisture in alpine meadows exhibits a consistent evaporation fractionation control pattern across temporal and spatial dimensions (Fig. 6). On the one hand, during summer (June–September), the soil water line (SWL) slope declines significantly, falling far below the local atmospheric water line (7.71), while the intercept value drops to its lowest point of − 20.93 ‰, indicating strong evaporative fractionation. In autumn, the slope and intercept rebound and nearly align with the atmospheric water line by October; critically, this rebound reflects infiltration and mixing of isotopically depleted new precipitation, rather than a reversal of the prior evaporation signal itself; on the other hand, the predominantly negative lc-excess histogram indicates widespread evaporative effects. Surface soils (0–10 cm) are ~ 1–12 ‰ more negative than 10–20 cm, consistent with stronger near-surface evapotranspiration. Under warming–humidification (ZWZY, ZWJY), the most negative seasonal values (≈−25 ‰) occur at the surface in August–September, evidencing amplified evaporation under high temperature and low humidity. Relative to the control deep layer, the warming group (ZWCK) shows lc-excess that first increases and then decreases, with an overall upward tendency. In October, lc-excess rises with autumn onset or post-storm conditions, reflecting precipitation-driven "dilution" (infiltration and mixing with resident soil water) rather than a reversal of evaporation per se. Spatial heterogeneity and preferential flow further magnify inter-layer and inter-treatment contrasts during peak summer (Sprenger et al., 2016; Gazis & Feng, 2004). Both sets of indicators point to the following: evaporation fractionation at the soil-atmosphere interface is the primary controlling process for summer isotope composition, with its intensity increasing with warming and surface moisture content and then sharply declining downward; rapid replenishment from new precipitation in autumn suppresses evaporation signals and reshapes the soil water isotope baseline. It is inferred that, under future warming and humidification scenarios, neglecting evaporation correction—i.e., the surface-evaporation fractionation correction applied to lc-excess values—will systematically overestimate the contribution of deep ice–snow water or groundwater to soil water; in source mixing or process models, the surface-deep lc-excess difference and warming-humidification interaction coefficients should be introduced separately to improve the

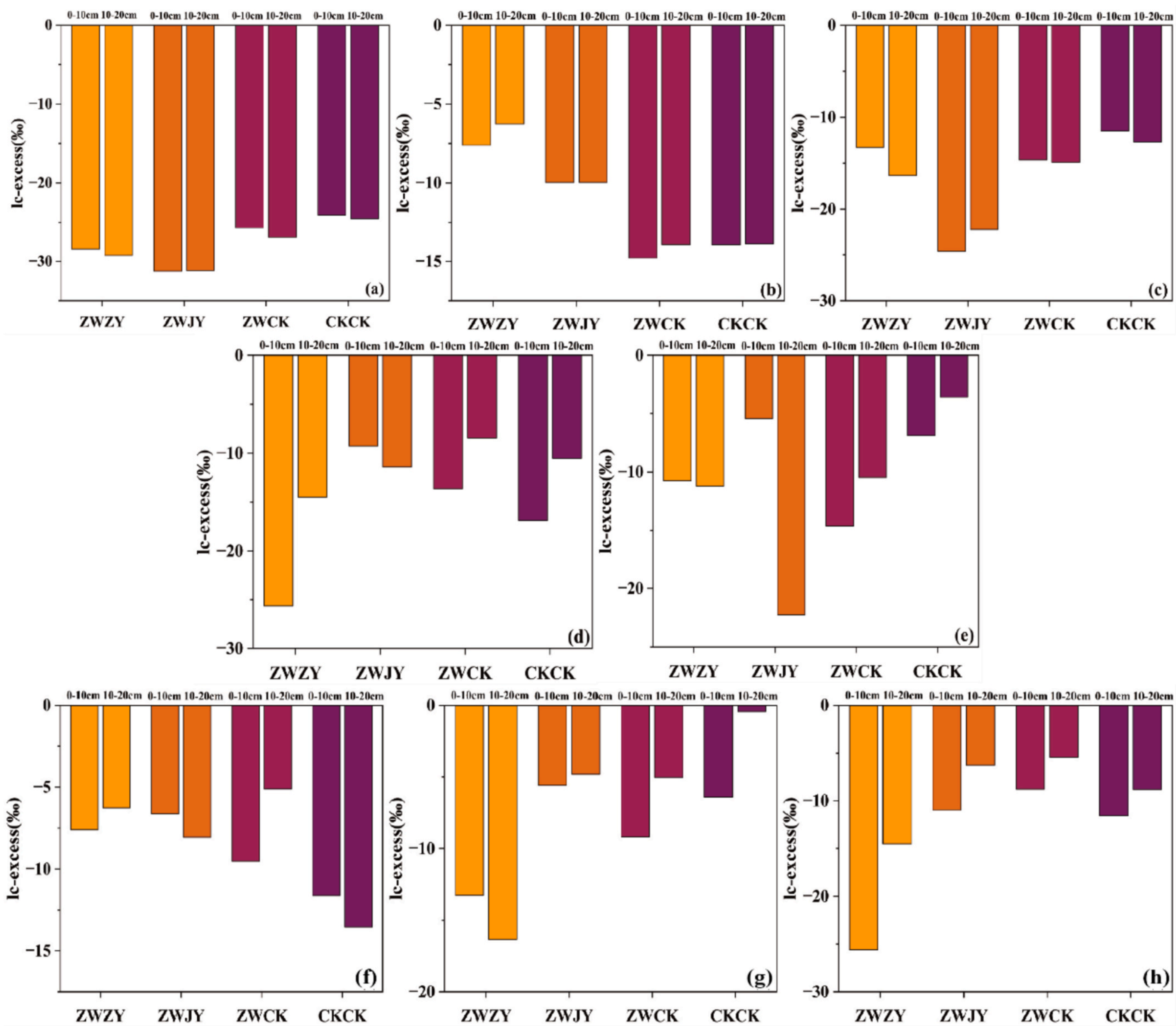

Fig. 6. Changes in IC-excess from June to October 2023 (a–e) and July to September 2024 (f–h), (ZWZY: warming + increased precipitation, ZWJY: warming + decreased precipitation, ZWCK: warming + control check, and CKCK: control check).

simulation accuracy of soil water supply pathways and water balance in high-altitude ecosystems.

Table 2 shows the monthly distribution pattern of soil water recharge in the deep layer (>10 cm) of alpine meadows between piston flow and preferential flow mechanisms during the 2023–2024 period, exhibiting a typical "piston dominance in early and late seasons—preferential flow activation in midsummer" double-peak structure. Specifically, the piston flow accounted for over 85 % (up to 87 %) in June–July and October, indicating that after snowmelt and into early summer, as well as during the autumn cooling phase, soil pores had not yet opened extensively to form interconnected pathways, and water movement was primarily slow, layered seepage, with deep-layer replenishment minimally affected by atmospheric precipitation fluctuations. In August–September, both years saw a sharp decline in piston flow and a significant increase in priority flow: in 2023 and 2024, the priority flow proportion rose synchronously to 38 % in August, and further increased to 48 % in September, indicating that concentrated heavy rainfall triggered rapid bypass flow in macropores or fractures, allowing new water pulses that had not undergone sufficient evaporation and fractionation to bypass the surface reservoir and reach the deep layer directly. The values for the two years were almost identical, suggesting that the region is controlled by stable monsoon rainfall events during midsummer to early autumn, and that the contribution of preferential flow exhibits significant inter-annual recurrence. Overall, when the priority flow ratio exceeds a threshold of approximately 35 % (Xiang et al.,2019; Wang et al., 2025), the renewal rate and source composition of deep soil water undergo a qualitative change, characterized by rapid recharge and isotopic "dilution," while deep water in other months is primarily sustained by

Table 2
Relative contributions of piston flow (PF) and preferential flow (pf) to deep soil water recharge.

Month	f_{PF}/%	f_{pf}/%
June	86	14
July	87	13
August	62	38
September	52	48
October	86	14

gradual piston seepage. This pattern provides direct parameter-based evidence for constructing groundwater recharge models considering seasonal dual mechanisms and assessing the potential impacts of warming and humidification on water redistribution in alpine meadows.

4.3. Quantitative analysis of soil water sources under warming

Projecting the average $\delta^{18}O$ and $\delta^{2}H$ values of soil water and the d-excess $\delta^{18}O$ isotope values into the "subsurface ice–precipitation–supra-permafrost water" three-terminal framework allows for a clear distinction of the mixed sources of shallow water bodies (Fig. 7). The d-excess $\delta^{18}O$ of surface soil water is approximately 8 ‰, significantly higher than that of subsurface ice (−18 ‰), but lower than precipitation (≈16 ‰) and supra-permafrost water (≈15 ‰), indicating that although meltwater from old ice provided isotopically "light end members," its contribution appears to be diluted by inputs from recent precipitation and supra-permafrost water characterized by high d-excess; however, the exact magnitude cannot be resolved without formal uncertainty propagation. Based on a simplified estimation using two-dimensional isotope mass conservation, precipitation accounts for approximately 55 %–65 % of soil water, supra-permafrost water contributes 25 %–35 %, and residual subglacial ice is less than 15 %. These results reveal that, under warming conditions, the deepening of supra-permafrost water and seasonal precipitation replenishment have become the dominant processes regulating the isotopic composition of surface soil water, while the direct influence of subglacial meltwater is being weakened by rapid hydrological cycling (Li et al., 2024a).

Fig. 8a–e compares the monthly contribution rates of the three water sources in the surface soil between the CKCK and the ZWCK from June to October 2023. It can be seen that warming significantly altered the water source structure and exhibited clear seasonal signals. Overall, precipitation remains the primary water source, but its proportion is generally higher in CKCK (46 %–61 %) than in ZWCK (29 %–55 %), indicating that warming has weakened the retention of new precipitation in the shallow layer. The contribution of supra-permafrost water to ZWCK increased markedly: in June and August it accounted for 51 % and 46 %, respectively—at least 7–18 percentage points higher than in CKCK—indicating that warming has accelerated the infiltration of supra-permafrost water into the surface layer. Although underground ice meltwater has the lowest overall proportion, it shows an upward trend in the later stages of high temperatures, reaching 26 % in September and 29 % in October, both higher than the corresponding periods in CKCK (23 % and 25 %), suggesting that the sustained melting triggered by warming provides additional "old water" sources for late-season soil water. In summary, mild warming significantly increased the mixing proportion of deep water sources in the near-surface water reservoir. Among the three candidate mechanisms—reduced precipitation retention, increased active-layer seepage, and release of meltwater from ground ice—only increased active-layer seepage is directly supported by the isotope data; reduced retention and meltwater release are presented as inferred processes consistent with site conditions and prior studies rather than demonstrated by the isotopes. These results highlight the sensitive response pathway of water circulation in high-altitude permafrost regions to climate warming (Bai et al., 2012).

The results of three-element water mixing in the surface soil of the warming zone (ZWCK) and the control zone (CKCK) from July to September 2024 show that warming significantly reshaped the water source structure and exhibited monthly reversal characteristics (Fig. 8f-h). In July, both zones were still dominated by seasonal precipitation (ZWCK 42 %, CKCK 44 %), but the proportion of supra-permafrost water in ZWCK had risen to 40 % (8 percentage points higher than CKCK), suggesting that warming had prematurely activated seepage from SPW (Qin et al., 2024). In August, monsoon precipitation intensified, with precipitation contributions in CKCK surging to 56 %, while ZWCK accounted for only 35 %, with the shortfall primarily filled by supra-permafrost water (47 %); simultaneously, the proportion of subsurface ice in CKCK plummeted to 6 %, corroborating that intense precipitation rapidly replenished and diluted the "old water" source (Li et al., 2024a). In September, as temperatures dropped and evapotranspiration weakened, ZWCK still relied primarily on supra-permafrost water (53 %), while CKCK underground ice rebounded to 30 %, indicating that the control area was re-supplied by deep meltwater in the late season, while the warming area, due to reduced water content and pore contraction, allowed new precipitation to easily penetrate along preferential flow to the upper active layer, weakening the direct influence of underground ice (Walvoord & Kurylyk, 2016). In summary, warming operates through a triple mechanism of "accelerating active layer deepening—shortening precipitation retention time—regulating the timing of subglacial ice meltwater release," ensuring that the contribution of supra-permafrost water to shallow reservoirs remains at a high level throughout the year (40–53 %). In contrast, the control area relies on new water during the precipitation peak period and compensatorily introduces subsoil ice meltwater in the late season, highlighting the high sensitivity and significant temporal differentiation of water circulation in permafrost regions to climate warming.

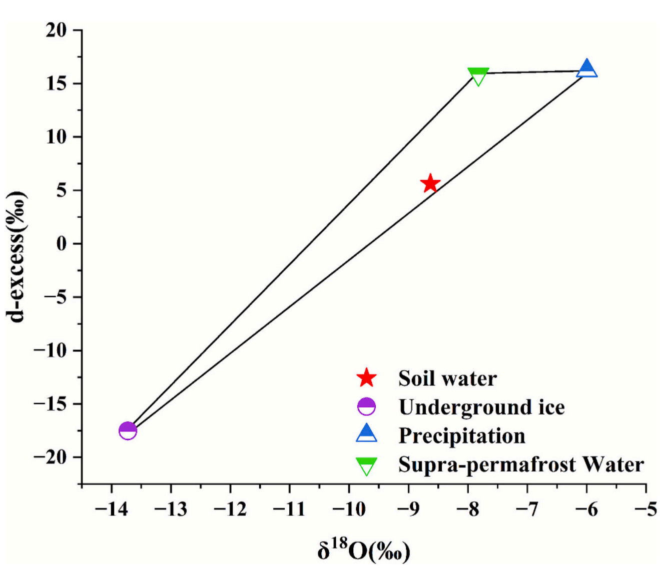

Fig. 7. Three-endmember mixing plots for soil water in $\delta^{18}O$–d-excess space.

4.4. Ecological restoration model of alpine meadows under warming conditions

The results showed significant changes in soil moisture content and in the fractional contributions of soil–water sources. These patterns are consistent with prior findings that experimental warming dries the near surface and steepens vertical moisture gradients via enhanced evapotranspiration (Wang and Wu, 2013; Weng et al., 2025), and with isotope-based mixing studies reporting seasonal shifts toward isotopically depleted melt inputs during monsoon and melt periods (Li et al., 2024a). For example, cold-region ecosystems are facing unprecedented vulnerability, and regional water resource management is also facing severe challenges (Gu et al., 2025). Permafrost degradation and the restructuring of the water-heat regime are altering soil moisture dynamics, energy allocation, and vegetation functions, necessitating systematic theoretical frameworks and management strategies to address these changes. Hu et al. (2024) quantified the primary factors controlling energy allocation and evapotranspiration in alpine meadows, noting that warming could increase the proportion of latent heat flux by 10 %–18 %; Li et al. (2024b) predicted that under the SSP5-8.5 scenario, the active layer thickness on the Qinghai-Tibet Plateau will increase by 39.6 %, coupled with melt depth-water redistribution. Warming effects manifest as multi-scale, multi-process coupled feedback: on one hand,

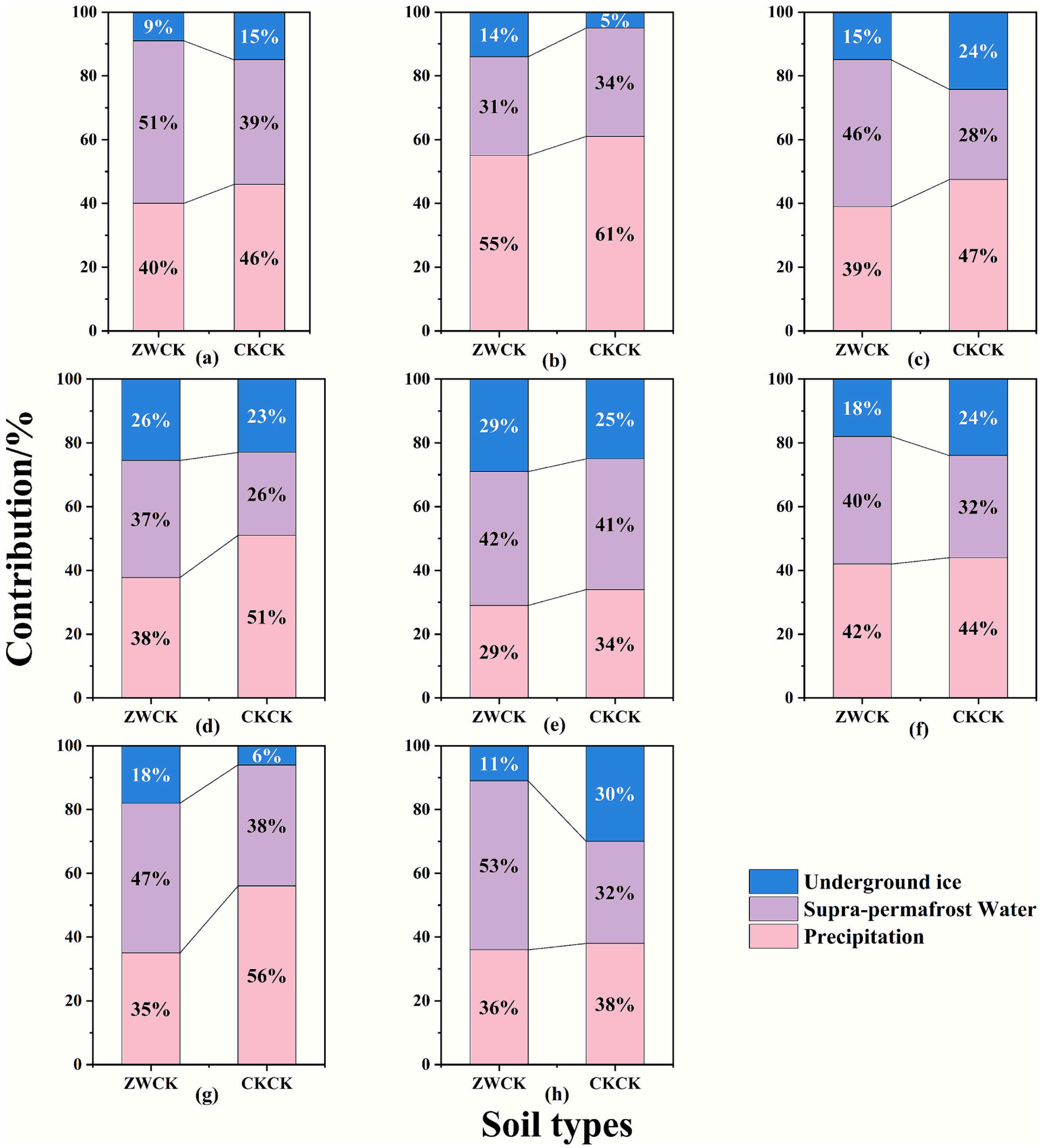

Fig. 8. Monthly soil water source contributions in CKCK vs. ZWCK for June–October 2023 (a-e) and July–September 2024 (f-h), (ZWCK: warming + control check, and CKCK: control check).

the melting of deeper permafrost and the deepening of the active layer increase the soil's instantaneous water storage capacity, but the resulting accelerated vertical water movement may lead to surface aridification and nutrient loss; on the other hand, the restructuring of soil moisture dynamics affects vegetation growth and distribution patterns, thereby weakening soil and water conservation functions, and alters regional water cycles through evapotranspiration and subsurface runoff processes, further disrupting ecosystem stability (Ahn S. R. et al., 2025). Given this, this study aims to comprehensively assess the impacts of climate warming and human activities on changes in soil moisture dynamics in cold regions, develop protection and restoration strategies suitable for cold regions, and provide scientific basis for the sustainable management of cold region ecosystems.

Under the warming scenario of alpine meadows, ecological restoration must address four key mechanisms: enhanced evapotranspiration, thickening of the active layer of permafrost, intensified water resource regulation pressures, and vegetation degradation. This study proposes corresponding ecological restoration recommendations and implements progressive interventions based on whether warming occurs (Fig. 9). ΔT denotes the temperature anomaly relative to the site-specific baseline, calculated as $\Delta T = T - T\_baseline$ (°C); positive values indicate warming and negative values indicate cooling. Once significant warming ($\Delta T > 0$ °C) is detected, interventions should first address the three stressors of enhanced evapotranspiration, thawing depth of permafrost, and water resource mismatch. This includes maintaining or replenishing grass mats and litter, covering high-water-consumption areas with paper-based

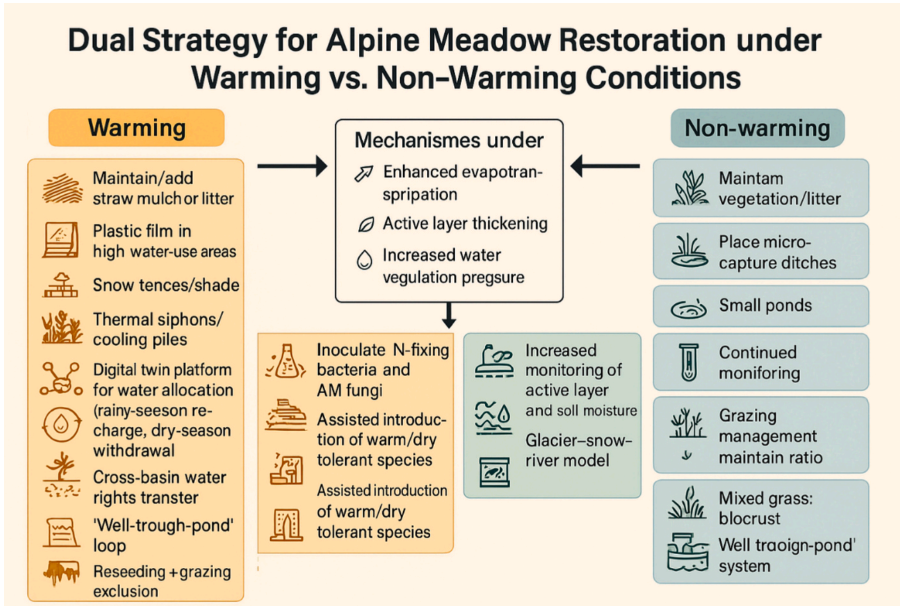

Fig. 9. Ecological restoration model of alpine meadows under warming and non-warming conditions.

mulch (Li et al., 2021; Jiang et al., 2025; Bi et al., 2021), gravel or sand mulching, and natural fiber geotextiles (Shao et al., 2014), installing windbreak snow fences and shade strips, and implementing key engineering measures such as heat sink pipes/ condensation piles, and other measures to simultaneously reduce surface-permafrost heat flux. Additionally, because isotopes diagnose source composition rather than recharge rates (Klaus & McDonnell, 2013; Kirchner, 2016; Clark & Fritz, 1997). Embedding quantified isotope metrics in a watershed digital twin yields two actionable operating rules. Flood season: when monitoring shows a higher share of "young water/new precipitation" and isotope signatures trending toward the meteoric line (Jasechko et al., 2016), prioritize local surface-water use and defer inter-basin transfers to protect supply and ecosystems (Yang et al., 2024; Brocca et al., 2024). Low-flow season: when isotope evidence indicates a larger fraction of "older/ baseflow-supported" water, tighten withdrawals and initiate inter-basin replenishment or water-rights transfers as needed (Pal et al., 2025; Yang et al., 2024). In practice, isotope-derived source shares and young-water thresholds can be mapped to scheduling triggers and allocation weights within the digital twin, helping rebalance water production and runoff. This approach converts process understanding into enforceable rules for water rights and use. For pastoral water management, promote a three-tiered circulation system of "wells-troughs-ponds." At the ecological restoration level, implement "enclosure and reseeding" combined with a rotational grazing system with a grazing-rest ratio $\geq$ 1:1, mixed planting of grasses and shrubs—biological soil crusting, large-scale introduction of deep-rooted drought-tolerant herbaceous plants (such as Stipa purpurea) combined with nitrogen-fixing bacteria and arbuscular mycorrhizal fungi inoculation, and terrace fields—biocarbon/ humic acid comprehensive improvement, supplemented by "auxiliary ex situ conservation" replanting of warm-dry adapted species, to achieve synergistic resilience enhancement of vegetation and soil; simultaneously, densify activity layers and soil moisture monitoring wells, combine glacier/snow-river dual models to predict agricultural and pastoral water demand, and implement zoned water restrictions to form a complete monitoring-warning-response closed-loop system. If the region has not yet experienced significant warming ($\Delta T \approx 0$ °C), the focus of management shifts to maintaining stability and preventing risks: continue to preserve existing vegetation and litter layers, lay straw mats in some areas to reduce evaporation from bare ground (Xue et al., 2025), and install controllable drainage ditches in low-lying areas and construct small reservoirs or rainwater collection facilities to cope with drought fluctuations; Regularly monitor permafrost active layers and soil temperature and humidity to establish an early warning system; Continue grassland management through enclosure plus reseeding, controlling livestock density, and promoting grass-shrub mixed planting and biological soil crusting technology (Ma et al., 2024); In terms of water resources, maintain precise dam scheduling and the "well-trough-pond" system to ensure ecological base flow and daily water use in pastoral areas (Wen et al., 2023). Through this dual-track strategy, rapid switching or overlapping responses can be achieved under the uncertainty of climate variability, effectively enhancing the soil and water conservation capacity and ecosystem resilience of alpine meadows.

5. Conclusions

Warming experiments have markedly reshaped soil–water dynamics and recharge pathways in the alpine meadows on the southern flank of the Qilian Mountains (Qinghai Province). Apart from the surface layer in the warming-plus-precipitation treatment (ZWZY), soil-moisture content in 2024 exceeded that in 2023 at every depth, whereas the warming-only treatment (ZWCK) intensified moisture loss across all horizons—underscoring the regulatory effect of additional rainfall under rising temperatures. Stable-isotope records (δ^2H, δ^{18}O) trace a consistent seasonal sequence: early-summer evaporative enrichment, mid-summer monsoonal dilution, and late-autumn re-enrichment. From July to September, the soil–water line exhibits a markedly lower slope than the local meteoric water line, indicating that intense summer rainfall infiltrates rapidly and dominates recharge; a rebound is observed in October. Statistical analysis confirms soil temperature as the primary driver of isotope variability, followed by precipitation; preliminary evidence suggests vegetation structure and pore architecture also play roles—an avenue for future study. Flow-path partitioning exhibits a clear "double-peak" pattern: piston flow prevails in June–July and October ($\geq$ 85 %), while heavy storms in August–September activate macropores and fractures, boosting preferential flow to 38–48 %. Three-end-member mixing reveals that supra-permafrost water now supplies 40–53 % of deep soil moisture, with subglacial ice relegated to a secondary role—showing that, under warming, the permafrost aquifer and seasonal precipitation jointly dominate recharge. Building on these insights, we propose a dual-track restoration strategy. During pronounced warming phases, priority should be given to surface-cover retention for cooling, digital-twin-guided precision water allocation, night-time irrigation, and a three-tier water-management system to curb surface–permafrost heat flux and mitigate water deficits. In non-warming phases, emphasis shifts to maintaining vegetative cover, implementing rational grazing, and establishing small-scale water-storage and regulation measures to bolster meadow resilience and maintain hydrological stability. Collectively, these measures offer a scientific basis and practical model for water-resource management and ecological restoration on the Qinghai–Tibet Plateau and in similar high-altitude regions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Ala-Aho, P., Soulsby, C., Pokrovsky, O.S., Kirpotin, S.N., Karlsson, J., Serikova, S., Vorobyev, S.N., Manasypov, R.M., Loiko, S., Tetzlaff, D., 2018. Using stable isotopes to assess surface water source dynamics and hydrological connectivity in a high-latitude wetland and permafrost influenced landscape. J. Hydrol. 556, 279–293. https://doi.org/10.1016/j.jhydrol.2017.11.024.

Allison, G.B., Hughes, M.W., 1983. The use of natural tracers as indicators of soil-water movement in a temperate semi-arid region. J. Hydrol. 60 (1–4), 157–173. https://doi.org/10.1016/0022-1694(83)90019-7.

Bai, W., Wang, G., Liu, G., 2012. Effects of elevated air temperatures on soil thermal and hydrologic processes in alpine meadows of the Qinghai–Tibet Plateau. J. Mt. Sci. 9, 229–239. https://doi.org/10.1007/s11629-012-2226-3.

Beven, K., Germann, P., 2013. Macropores and water flow in soils revisited. Water Resour. Res. 49 (6), 3071–3092. https://doi.org/10.1002/wrcr.20156.

Bi, S., Pan, H., Barinelli, V., Eriksen, B., Ruiz, S., Sobkowicz, M.J., 2021. Biodegradable polyester coated mulch paper for controlled release of fertilizer. J. Clean. Prod. 294, 126348. https://doi.org/10.1016/j.jclepro.2021.126348.

Brocca, L., Barbetta, S., Camici, S., Ciabatta, L., Dari, J., Filippucci, P., Massari, C., Modanesi, S., Tarpanelli, A., Bonaccorsi, B., Mosaffa, H., Wagner, W., Vreugdenhil, M., Quast, R., Alfieri, L., Gabellani, S., Avanzi, F., Rains, D., Miralles, D. G., Fernandez, D., 2024. A digital twin of the terrestrial water cycle: A glimpse into the future through high-resolution Earth observations. Frontiers in Science 1, 1190191. https://doi.org/10.3389/fsci.2023.1190191.

Cao, X., Wang, Y., Wu, P., Zhao, X., Wang, J., 2015. An evaluation of the water utilization and grain production of irrigated and rain-fed croplands in China. Sci. Total Environ. 529, 10–20. https://doi.org/10.1016/j.scitotenv.2015.05.050.

Chen, H., Zhu, Q., Peng, C., et al., 2013. The impacts of climate change and human activities on biogeochemical cycles on the Qinghai-Tibetan Plateau. Glob. Chang. Biol. 19 (10), 2940–2955. https://doi.org/10.1111/gcb.12277.

Cheng, G., Jin, H., 2013. Permafrost and groundwater on the Qinghai-Tibet Plateau and in Northeast China. Hydrgeol. J. 21 (1), 5–23. https://doi.org/10.1007/s10040-012-0927-2.

Clark, I.D., Fritz, P., 1997. *Environmental isotopes in hydrogeology*. CRC Press. https://doi.org/10.1201/9781482242911.

Cong, P., Tan, H., Shi, D., et al., 2023. Understanding the soil and plant water hydrological cycle in a typical wetland under harsh natural conditions in the Shigatse area of the Tibetan Plateau. Hydrol. Process. 37 (6), e14921. https://doi.org/10.1002/hyp.14921.

Correa, A., Windhorst, D., Tetzlaff, D., Crespo, P., Célleri, R., Feyen, J., Breuer, L., 2017. Temporal dynamics in dominant runoff sources and flow paths in the Andean Páramo. Water Resour. Res. 53 (7), 5998–6017. https://doi.org/10.1002/2016WR020187.

Dawson, T.E., Ehleringer, J.R., 1991. Streamside trees that do not use stream water. Nature 350 (6316), 335–337. https://doi.org/10.1038/350335a0.

Du, F., Li, Z., Gui, J., et al., 2024. Mechanisms of suprapermafrost groundwater-recharge streamflow in alpine permafrost regions: Insights from young-water-fraction analysis. Water Resour. Res. 60, e2023WR034921. https://doi.org/10.1029/2024WR037329.

Gangcha County Annals Compilation Committee. (1998). *Gangcha County Annals* (pp. 95–97). Shaanxi People's Publishing House.

Gazis, C., Feng, X., 2004. A stable isotope study of soil water: Evidence for mixing and preferential flow paths. Geoderma 119 (1–2), 97–111. https://doi.org/10.1016/S0016-7061(03)00243-X.

Geris, J., Tetzlaff, D., McDonnell, J., et al., 2015. The relative role of soil type and tree cover on water storage and transmission in northern headwater catchments. Hydrol. Process. 29 (7), 1844–1860. https://doi.org/10.1002/hyp.10289.

Gu, B., Zhou, S., Yu, B., Findell, K.L., Lintner, B.R., 2025. Multifaceted changes in water availability with a warmer climate. npj Clim. Atmos. Sci. 8, 31. https://doi.org/10.1038/s41612-025-00913-4.

Gui, J., Li, Z., Du, F., Xue, J., Zhang, B., Wang, D., 2023. Source of soil water in cold regions based on stable isotope tracers. Global Ecol. Conserv. 45, e02518. https://doi.org/10.1016/j.gecco.2023.e02518.

Hooper, R.P., 2003. Diagnostic tools for mixing models of stream-water chemistry. Water Resour. Res. 39 (3), 1055. https://doi.org/10.1029/2002WR001528.

Hu, Z.Y., Li, X.Q., Chen, W., 2024. Quantifying alpine meadow energy partitioning under experimental warming. Agric. For. Meteorol. 343, 109575. https://doi.org/10.1016/j.agrformet.2024.109575.

Huang, T., Ma, B., Pang, Z., Li, Z., Li, Z., Long, Y., 2020. How does precipitation recharge groundwater in loess aquifers? Evidence from multiple environmental tracers. J. Hydrol. 583, 124532. https://doi.org/10.1016/j.jhydrol.2020.124532.

Jasechko, S., Kirchner, J.W., Welker, J.M., McDonnell, J.J., 2016. Substantial proportion of global streamflow less than three months old. Nat. Geosci. 9 (2), 126–129. https://doi.org/10.1038/ngeo2636.

Jiang, C., Wu, J., Wang, H., Tao, Y., Lu, J., Hu, J., Wang, H., 2025. Multiple cross-linked cellulosic paper-based waterproof and biodegradable mulch film for "green" agriculture. Int. J. Biol. Macromol. 284, 137977. https://doi.org/10.1016/j.ijbiomac.2024.137977.

Jiao, L., An, W., Li, Z., Gao, G., Wang, C., 2020. Regional variation in soil water and vegetation characteristics in the Chinese Loess Plateau. Ecol. Ind. 115, 106399. https://doi.org/10.1016/j.ecolind.2020.106399.

Kirchner, J.W., 2016. Aggregation in environmental systems—Part 1: Seasonal tracer cycles quantify young water fractions, but not mean transit times, in spatially heterogeneous catchments. Hydrol. Earth Syst. Sci. 20 (1), 279–297. https://doi.org/10.5194/hess-20-279-2016.

Klaus, J., McDonnell, J.J., 2013. Hydrograph separation using stable isotopes: Review and evaluation. J. Hydrol. 505, 47–64. https://doi.org/10.1016/j.jhydrol.2013.09.006.

Li, Z., Gui, J., Cui, Q., et al., 2024a. Soil water sources and their implications for vegetation restoration in the Three-Rivers Headwater Region during different ablation periods. Hydrol. Earth Syst. Sci. 28 (3), 719–734. https://doi.org/10.5194/hess-28-719-2024.

Li, A., Zhang, J., Ren, S., Zhang, Y., Zhang, F., 2021. Research progress on preparation and field application of paper mulch. Environ. Technol. Innovation 24, 101949. https://doi.org/10.1016/j.eti.2021.101949.

Li, Y., Li, X., Cui, B., Peng, H., Yi, W., 2010. Trend of streamflow in Lake Qinghai basin during the past 50 years(1956-2007)-Take Buha River and Shaliu River for examples. Journal of Lake Sciences 22 (5), 757–766. https://doi.org/10.18307/2010.0519.

Li, Z., Su, B., Huang, J., Havea, P.H., Xu, R., Jing, C., Jiang, T., 2024b. Projected changes of Active layer thickness over permafrost under 1.5∼ 5.0° C climate warming on the Qinghai-Tibet Plateau. EGUsphere 2024, 1–20. https://doi.org/10.5194/egusphere-2024-1169.

Li, Z.X., Feng, Q., Wang, Q.J., Song, Y., Cheng, A., 2016a. Contribution from frozen-soil meltwater to runoff in an inland river basin under water scarcity by isotopic tracing in north-western China. Global Planet. Change 136, 41–51. https://doi.org/10.1016/j.gloplacha.2015.12.002.

Li, Z.X., Feng, Q., Wang, Q.J., Song, Y., Li, H., Li, Y., 2016b. The influence of the shrinking cryosphere and strengthening evapotranspiration on hydrologic processes in a cold basin, Qilian Mountains. Global Planet. Change 144, 119–128. https://doi.org/10.1016/j.gloplacha.2016.06.017.

Li, Z., Gui, J., Cui, Q., Xue, J., Du, F., Si, L., 2024c. Soil water sources and their implications for vegetation restoration in the Three-Rivers Headwater Region during different ablation periods. Hydrol. Earth Syst. Sci. 28, 719–734. https://doi.org/10.5194/hess-28-719-2024.

Ma, Y., Liu, Y., Rodrigo-Comino, J., López-Vicente, M., Wu, G.-L., 2024. Mixed-cultivation grasslands enhance runoff generation and reduce soil loss in the restoration of degraded alpine hillsides. Hydrol. Earth Syst. Sci. 28 (16), 3947–3961. https://doi.org/10.5194/hess-28-3947-2024.

McDonnell, J.J., Beven, K., 2014. Debates—The future of hydrological sciences: A (common) path forward? A call to action aimed at understanding velocities, celerities and residence-time distributions of the headwater hydrograph. Water Resour. Res. 50 (6), 5342–5350. https://doi.org/10.1002/2013WR015141.

Pal, D., Marttila, H., Ala-Aho, P., Lotsari, E., Ronkanen, A.-K., Gonzales-Inca, C., Croghan, D., Korppoo, M., Kamari, M., van Rooijen, E., Blåfield, L., Silander, J., Baubekova, A., Bhattacharjee, J., Haghighi, A.T., Uvo, C.B., Kaartinen, H., Rasti, M., Klöve, B., Alho, P., 2025. Blueprint conceptualization for a river basin's digital twin. Hydrol. Res. 56 (3), 197–212. https://doi.org/10.2166/nh.2025.111.

Penna, D., Hopp, L., Scandellari, F., Allen, S.T., Benettin, P., Beyer, M., Geris, J., Klaus, J., Marshall, J.D., Schwendenmann, L., Volkmann, T.H.M., von Freyberg, J., Amin, A., Ceperley, N., Engel, M., Frentress, J., Giambastiani, Y., McDonnell, J.J., Kirchner, J.

W., 2018. Ideas and perspectives: Tracing terrestrial-ecosystem water fluxes using hydrogen and oxygen stable isotopes—Challenges and opportunities from an interdisciplinary perspective. Biogeosciences 15 (21), 6399–6415. https://doi.org/10.5194/bg-15-6399-2018.

Penna, D., van Meerveld, H.J., Zuecco, G., Dalla Fontana, G., Borga, M., 2016. Hydrological response of an Alpine catchment to rainfall and snowmelt events. J. Hydrol. 537, 382–397. https://doi.org/10.1016/j.jhydrol.2016.03.040.

Pyschik, J., & Weiler, M. (2025). Detecting the occurrence of preferential flow in soils with stable water isotopes (EGUsphere preprint). doi: 10.5194/egusphere-2025-2411.

Qin, J., Ding, Y., Shi, F., Cui, J., Chang, Y., Han, T., Zhao, Q., 2024. Links between seasonal suprapermafrost groundwater, the hydrothermal change of the active layer, and river runoff in alpine permafrost watersheds. Hydrol. Earth Syst. Sci. 28, 973–987. https://doi.org/10.5194/hess-28-973-2024.

Shao, Q., Gu, W., Dai, Q.Y., Makoto, S., Liu, Y., 2014. Effectiveness of geotextile mulches for slope restoration in semi-arid northern China. Catena 116, 1–9. https://doi.org/10.1016/j.catena.2013.12.006.

Sprenger, M., Leistert, H., Gimbel, K., Weiler, M., 2016. Illuminating hydrological processes at the soil–vegetation–atmosphere interface with water stable isotopes. Rev. Geophys. 54 (3), 674–704. https://doi.org/10.1002/2015RG000515.

Sprenger, M., Tetzlaff, D., Soulsby, C., 2017. Soil-water stable isotopes reveal evaporation dynamics at the soil–plant–atmosphere interface of the critical zone. Hydrol. Earth Syst. Sci. 21 (7), 3839–3858. https://doi.org/10.5194/hess-21-3839-2017.

Throckmorton, H.M., Newman, B.D., Heikoop, J.M., Graham, D.E., O'Malley, D., Vesselinov, V.V., Young, J., Wilson, C.J., Wullschleger, S.D., 2016. Active layer hydrology in an Arctic tundra ecosystem: Quantifying water sources and cycling using water stable isotopes. Hydrol. Process. 30 (26), 4972–4986.

Walvoord, M.A., Kurylyk, B.L., 2016. Hydrologic impacts of thawing permafrost—A review. Vadose Zone J. 15 (6), vzj2016.01.0010. https://doi.org/10.2136/vzj2016.01.0010.

Wang, J., Wu, Q., 2013. Impact of experimental warming on soil temperature and moisture of the shallow active layer of wet meadows on the Qinghai–Tibet Plateau. Cold Reg. Sci. Technol. 90–91, 1–8. https://doi.org/10.1016/j.coldregions.2013.03.005.

Wang, Z., Jia, W., Zhang, Y., Lan, X., Yu, Z., Luo, H., Chou, L., 2025. The recharge and infiltration of soil water in the subalpine shrub zone of the eastern Qilian Mountains in China. Environ. Earth Sci. 84, 290. https://doi.org/10.1007/s12665-025-12312-2.

Weiler, M., McDonnell, J.J., 2004. Virtual experiments: A new approach for improving process conceptualization in hillslope hydrology. J. Hydrol. 285 (1–4), 3–18. https://doi.org/10.1016/j.jhydrol.2003.07.015.

Wen, X., Shao, H., Wang, Y., Lv, L., Xian, W., Shao, Q., Shu, Y., Yin, Z., Liu, S., Qi, J., 2023. Assessment of the spatiotemporal impact of water conservation on the Qinghai–Tibet Plateau. Remote Sens. (Basel) 15 (12), 3175. https://doi.org/10.3390/rs15123175.

Weng, B., Yan, S., Fang, H., et al., 2025. Warming increases root water uptake on the Qinghai-Tibet Plateau via changes in alpine meadows root tips. Agric Water Manag 312 (C). https://doi.org/10.1016/j.agwat.2025.109438.

Xiang, W., Si, B.C., Biswas, A., Li, Z., 2019. Quantifying dual recharge mechanisms in deep unsaturated zone of Chinese Loess Plateau using stable isotopes. Geoderma 337, 773–781. https://doi.org/10.1016/j.geoderma.2018.10.006.

Xue, G., Tian, L., Zhao, J., 2025. Effects of simulated warming and litter removal on structure and function of semi-humid alpine grassland in the Qinghai–Tibet Plateau. Front. Plant Sci. 16, 1567414. https://doi.org/10.3389/fpls.2025.1567414.

Yang, Y., Cao, S., Cao, G., et al., 2020. Recharge characteristics of shallow groundwater in different periods of Shaliu River Basin around the Qinghai Lake. Arid Land Geography 43 (3), 633–643. https://doi.org/10.12118/j.issn.1000-6060.2020.03.09.

Yang, Y., Xie, C., Fan, Z., Xu, Z., Melville, B.W., Liu, G., Hong, L., 2024. Digital twinning of river basins towards full-scale, sustainable and equitable water management and disaster mitigation. npj. Nat. Hazards 1, 43. https://doi.org/10.1038/s44304-024-00047-2.

Yao, T., Thompson, L.G., Mosbrugger, V., Zhang, F., Ma, Y., Luo, T., Xu, B., Yang, X., Joswiak, D.R., Wang, W., Joswiak, M.E., Devkota, L.P., Tayal, S., Jilani, R., Fayziev, R., 2012. Third Pole environment (TPE). Environmental Development 3, 52–64. https://doi.org/10.1016/j.envdev.2012.04.002.

Yao, T., Xue, Y., Chen, D., Thompson, L.G., 2019. Recent Third Pole rapid warming accompanies cryospheric melt and water-cycle intensification, and interactions between monsoon and environment: A multidisciplinary approach with observations, modeling and analysis. Bull. Am. Meteorol. Soc. 100 (3), 423–444. https://doi.org/10.1175/BAMS-D-17-0057.1.

Zhong, X., Zhang, M., Zhang, Y., Wang, J., Liu, Z., Gu, L., 2023. Soil-water infiltration process in north and south mountains of Lanzhou City based on stable isotope. Arid Zone Res. 40 (11), 1744–1753. https://doi.org/10.13866/j.azr.2023.11.04.

Zimmermann, U., Münnich, K.O., Roether, W., Kreutz, W., Schubach, K., Siegel, O., 1966. Tracers determine movement of soil moisture and evapotranspiration. Science 152 (3720), 346–347. https://doi.org/10.1126/science.152.3720.346.

Zong, N., Duan, C., Geng, S.-B., 2018. Effects of warming and nitrogen addition on community production and biomass allocation in an alpine meadow. Chin. J. Appl. Ecol. 29 (1), 59–67. https://doi.org/10.13287/j.1001-9332.201801.006.

Zongjie, L., Bin, X., Xiaoying, L., Zongxing, L., Qi, F., Dongpeng, W., Fang, L., 2025. Infiltration mechanism and source of soil water in alpine meadows based on stable isotope tracing. Geoderma 455, 117224. https://doi.org/10.1016/j.geoderma.2024.117224.